

HYOSEO LEE

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TECHNICAL SKILLS

Languages : Python, C, C++, TypeScript, SQL, HTML/CSS

Frameworks : NumPy, Pandas, Matplotlib, FastAPI, React, Astro

Tools & Platforms : AWS, Docker, Kubernetes, Postman, GitHub Actions, Ubuntu, Stripe

Databases & Services : PostgreSQL, LLM API, Supabase, Redis, Vercel

EXPERIENCE

Squad Leader & Fuel Supply Specialist

Gyeonggi, South Korea

Republic of Korea Army (ROKA) | Mandatory Military Service

Oct 2024 – Apr 2026

- Directed fuel logistics and supply chain operations for the battalion, maintaining 100% operational readiness with zero safety incidents over an 18-month deployment.
- Led, trained, and mentored a 14-soldier squad, optimizing daily operational workflows and conducting routine performance evaluations to ensure maximum efficiency.

”AI Developer (Co-op) – AI Stream

Remote

Microsoft-mentored Co-op Program

May 2024 – Aug 2024

- Architected and deployed an end-to-end Azure AI pipeline for a medical diagnosis chatbot, significantly improving query processing speed and diagnostic accuracy.
- Secured AZ-900 and AI-900 certifications, validating practical expertise in building and managing scalable cloud architectures and core AI services.

PROJECTS

Waterloo Works AI Assistant | *TypeScript, Python, React, FastAPI, PostgreSQL, Gemini API* | May 2026

- Scaled a Chrome extension into a monetized SaaS product, engineering an automated scoring engine using the **Gemini API** that injects fit scores and reduces manual job screening time by over **80%**.
- Built a **FastAPI** backend on DigitalOcean with **Microsoft Entra SSO** and a **Stripe** credit system, paired with a seamless **React/TypeScript** frontend bypassing complex authentication barriers.

Macro Dashboard | *React, FastAPI, PostgreSQL, Docker, DigitalOcean* |

Apr–May 2026

- Designed a production-grade financial terminal with a customizable **React UI**, featuring drag-and-drop widgets, real-time sparklines, and aggregated RSS news from **22 global sources**.
- Automated infrastructure using **Docker Compose** on **DigitalOcean** with a Caddy reverse proxy for auto-HTTPS, utilizing background schedulers for **real-time data synchronization**.

CxC Hackathon — Wildfire Analysis | *Python, Pandas, Matplotlib, Scikit-Learn* |

Feb–Mar 2024

- Analyzed **22,000+ Alberta wildfire records** and engineered a **Random Forest model** that achieved over **90% prediction accuracy** in forecasting burn sizes.
- Won the **MEF Award** out of dozens of competing teams for demonstrating profound analytical insight into the environmental and economic impact on **indigenous communities**.

Dchup — Flashcard Web App | *React, FastAPI, AWS (EC2, RDS), PostgreSQL, Netlify* |

Mar–Jul 2024

- Architected a full-stack learning platform, developing a **Python FastAPI** backend and a dynamic **React** frontend handling secure user authentication and flashcard CRUD operations.
- Engineered robust **CI/CD pipelines** to automate deployments, hosting the backend on **AWS EC2** with an RDS database and the frontend on **Netlify** to ensure high availability.

EDUCATION / CERTIFICATIONS

University of Waterloo

Waterloo, ON

Bachelor of Computer Science, Honours, Co-operative Program (92.33 cumulative GPA)

Sep 2023 – Present

Microsoft Certified: Azure Fundamentals (AZ-900 & AI-900)

Remote

Core cloud architectures and AI services

Jun 2024